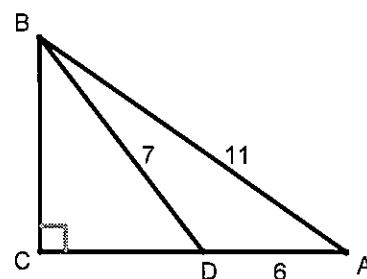


1. Let $k = \sin\left(\frac{2}{3}\pi\right)$ and $w = \cot\left(\frac{5}{6}\pi\right)$. Determine the exact product (kw) .
2. $\text{Arc sin } x$ represents the inverse sine function. Determine the exact value of $\text{Arc sin}\left(\cos\left(\frac{7}{6}\pi\right)\right)$.
3. Let $f(x) = 2x^2 - 5x - 9$. Then $\lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right) = kx^2 + wx + p$. Determine the value of $(k + w + p)$.
4. A particular triangle has sides of length 20 and 17 and an included angle between those two sides of measure 52° . Determine the area of this triangle. Express your answer as a decimal rounded to the nearest thousandth.
5. Let $k = \sum_{n=0}^{\infty} (18) \left(\frac{17}{20} \right)^n$. Determine the exact value of k .

6. $\triangle ABC$ is a right triangle with right angle at C . $AB = 11$, $BD = 7$, and $DA = 6$. Determine the length of $\overline{BC}$. Express your answer as a decimal rounded to the nearest thousandth.



7. Let matrices $A = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 7 & 3 \\ 3 & 2 \end{bmatrix}$. Determine the value of $\det(A \times B)$ (the determinant of the product of matrices A and B .)